



An-Najah National University

Shield Gates

Secure Email System with Mail Gateway Filtering

Technical Documentation

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Appendix-A

The features are extracted from an email so they can be used in machine learning (for example, phishing detection or spam classification). Each feature describes a property of the email header, subject, body, links, keywords, or attachments.

1. Basic Dataset Information

Feature	Meaning
ow_index	Row index of the email in the dataset
data_source	The dataset source (example: Enron dataset)
label	Email classification (usually 0 = legitimate , 1 = phishing/spam)
label_source	Where the label came from (manual label, dataset label, etc.)

2. Email Header Features

Feature	Meaning
has_from	An email contains a From: field
has_to	Email contains a To: field
has_cc	Email contains CC: recipients
has_bcc	Email contains BCC: recipients
has_subject	Email has a subject line
has_date	Email has a Date: field
has_message_id	Email has a Message-ID header
has_reply_to	Email has a Reply-To: address
has_return_path	Email has a Return-Path header
has_dkim	Email has DKIM authentication signature

3. Subject Line Features

Feature	Meaning
subject_length	Number of characters in the subject
subject_word_count	Number of words in the subject
subject_has_reply	Contains "Re:"
subject_has_fwd	Contains "Fwd:"
subject_has_urgent	Contains word "urgent"
subject_has_alert	Contains word "alert"
subject_has_verify	Contains "verify"
subject_all_caps	Subject written entirely in CAPITAL letters
subject_caps_ratio	Ratio of uppercase letters
subject_has_numbers	Subject contains digits
subject_has_special	Subject contains special characters
subject_exclamation	Contains !
subject_question	Contains ?
subject_money	Contains currency symbols

4. Body Size Features

Feature	Meaning
body_length	Number of characters in body
body_word_count	Number of words
body_line_count	Number of lines
body_paragraph_count	Number of paragraphs

5. Character Statistics

Feature	Meaning
uppercase_count	Number of uppercase letters
lowercase_count	Number of lowercase letters
digit_count	Number of numbers
space_count	Number of spaces
punctuation_count	Number of punctuation marks
special_char_count	Special characters (@ # \$ % &)

6. Character Ratios

Feature	Meaning
uppercase_ratio	Uppercase / total characters
lowercase_ratio	Lowercase / total characters
digit_ratio	Numbers / total characters
space_ratio	Spaces / total characters
punctuation_ratio	Punctuation ratio
special_char_ratio	Special characters ratio

7. Word Statistics

Feature	Meaning
avg_word_length	Average length of words
unique_word_count	Number of unique words
unique_word_ratio	Unique words / total words

8. URL / Link Features

Feature	Meaning
url_count	Number of URLs in the email
unique_url_count	Number of different URLs
url_avg_length	Average URL length
url_max_length	Longest URL
url_min_length	Shortest URL
url_has_ip	URL uses IP instead of domain
url_has_port	URL specifies port number
url_has_https	Uses HTTPS
url_has_http	Uses HTTP
url_count_https	Number of HTTPS links
url_count_http	Number of HTTP links
url_suspicious_tlds	Suspicious domain extensions (.ru, .tk, etc.)
url_has_subdomains	Contains many subdomains
url_max_dots	Maximum number of dots in URL
url_avg_slashes	Average number of / characters
url_has_at_symbol	Contains @ in URL
url_has_double_slash	Contains // in path
url_has_hyphen	Contains -
url_has_underscore	Contains _
url_has_percent_encoding	Contains encoded characters (%20, %2F)

9. Contact Information Detection

Feature	Meaning
email_in_body_count	Number of email addresses in body

unique_email_in_body_count	Unique email addresses
phone_count	Phone numbers detected
ip_address_count	IP addresses in email

10. Money Symbols

Feature	Meaning
dollar_sign_count	\$ count
euro_sign_count	€ count
pound_sign_count	£ count
yen_sign_count	¥ count
total_money_symbols	Total currency symbols

11. Punctuation Indicators

Feature	Meaning
exclamation_count	Number of !
question_count	Number of ?
exclamation_ratio	Ratio of !
question_ratio	Ratio of ?

12. HTML Content Features

Feature	Meaning
has_html_tags	Email contains HTML
html_tag_count	Number of HTML tags

13. Phishing Keyword Detection

Feature	Meaning
keyword_urgent	Word "urgent" appears
keyword_verify	"verify"
keyword_account	"account"
keyword_bank	"bank"
keyword_paypal	"paypal"
keyword_suspended	"suspended"
keyword_click	"click"
keyword_login	"login"
keyword_password	"password"
keyword_credit	"credit"
keyword_security	"security"
keyword_confirm	"confirm"

14. Keyword Statistics

Feature	Meaning
total_phishing_keywords	Total number of phishing keywords found
unique_phishing_keywords	Number of unique phishing keywords

15. Psychological Manipulation Scores

Feature	Meaning
urgency_score	Measures urgency language
fear_score	Measures fear-based language
curiosity_score	Measures curiosity bait

16. HTML Attack Indicators

Feature	Meaning
has_html_entities	Contains encoded HTML characters
html_entity_count	Number of HTML entities
has_javascript	JavaScript code detected
has_onclick	onclick event detected
has_onload	onload event detected
has_form	HTML form present
has_input	Input fields present

17. Attachment Features

Feature	Meaning
attachment_count	Number of attachments
has_executable_attachment	.exe, .bat, etc.
has_archive_attachment	.zip, .rar
has_pdf_attachment	.pdf
has_document_attachment	.doc, .docx
has_image_attachment	.jpg, .png